# NEA analysis - Hospital patient management software

## 3.1.1 Problem Identification:

### 3.1.1 a) What is the problem?:

In the modern age, the use of technology to improve the quality of life in fields like science and healthcare has become increasingly more popular in the 21st century. The National Library of Medicine has claimed that:

" Introduction of ICT in a hospital affects the operation of the organisation, health care delivery, and patient outcomes while offering potential benefits in improved patient safety, reduced hospital costs, and increased efficiency and effectiveness of medical care. "

And this has led me to produce this project. My programming project will be a hospital patient management system which will simplify hospital care for doctors, nurses and surgeons. I have chosen this topic for my project because the use of technology in fields such as medicine is becoming increasingly common in our modern society. However, after consulting stakeholders (my computing teachers and my family, who are mostly software engineers). They have all argued that the use of automated computer systems is vital and significant in the field of hospital care, where immediate and up-to-date information is crucial to ensure patient health. However, these health systems currently lack the up-to-date innovation and user-friendliness to allow nurses to view records in a minimalist manner. Furthermore, hospital staff and nurses face numerous challenges when it comes to storing patient records and viewing graphical information, such as X-rays, on an inefficient and non-user-friendly computer system. In conclusion, my computer system will be designed to cater to the needs of staff, allowing them to manage patient records efficiently, view injuries and x-rays, and ensure they have the necessary tools to perform their jobs effectively.

### 3.1.1 b) Why is this problem amenable to a computational approach? :

Many features of this hospital system lend themselves to a computational approach, and so here are a few features:

| Feature | Explanation | Manual | Computer based | Justification |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| Viewing & adding patient records | This is a feature that will allow staff to view and add patient information, in the form of "patient records", where a group of records is known as an "EHR". This EHR system enables staff to view, edit, add and delete patient information (e.g. symptoms, medication, medical conditions, etc.) | | ✓ | This is computer-based as adding patient records manually may lead to incorrect or inconsistent information on patients. Therefore, viewing & entering data on a computer system could reduce the chance of errors, compared to manual data entry, which carries higher risks. |
| Managing hospital rooms | This feature will enable hospital staff to view whether surgery rooms are available (unbooked) and unavailable (booked), to avoid booking clashes between staff. This feature will also enable staff to book surgery rooms for certain patients & for a set duration | | ✓ | While it is possible for staff to manually track which rooms are occupied, this can become confusing & error-prone, especially if a hospital is at peak times & is really busy. A computerised system can prevent conflicts when double-booking rooms, as well as provide quick searches for available rooms. |
| Storing patient data securely | Hospitals store highly sensitive data about their patients such as legal and medical information. Therefore, it is an obligation for hospitals under the Data Protection Act, to store this information in a safe and secure manner, that prevents unauthorised access into the system. | | ✓ | Computers are well-suited for large data storage as digital records cannot be easily lost or damaged, unlike paper files. Furthermore, patient records can be retrieved immediately, & hashing and authentication can be used to protect sensitive medical records from unauthorised access. |
| Automating medication reminders | Hospital staff face many pressures and challenges in their day-to-day jobs. This means that there is a high chance of staff forgetting certain tasks e.g. treating certain patients at set times throughout the day. This feature reduces that chance by providing automated alerts to staff, reminding them to treat certain patients at that time. | | ✓ | While this is possible for nurses to write down medication times for each patient, this will likely be time-consuming & there is a high risk of staff overlooking these tasks. Whereas computers can automatically generate alerts for medicine at the correct times, thus increasing reliability. |
| Viewing medical images (e.g., X-rays) | This feature allows staff to visualise patient injuries by offering sub-features such as CT scans, X-ray scans, & MRI scans. These types of medical imaging, will simplify diagnosing patients with certain diseases & see internal structures of the body | | ✓ | Paper records cannot handle digital high-resolution images, and printing can also reduce clarity. And so, a computerised system will allow hospital staff to view graphical details, e.g. X-rays and injury details, in a clearer and more user-friendly manner. |
| Allocating staff & resources | This feature allows staff to check whether or not their colleagues and co-workers are currently busy with servicing certain patients or not e.g. on break. Therefore, this allows them to communicate to members of staff, potentially asking for help on certain duties. E.g., preparing surgery equipment. | ✓ | | While a computer system can do this, resource management and staff allocation (nurses, doctors, surgeons) will likely be easier for staff members to do to each other, i.e. distributing medical tasks to each other through group meetings. While this is better done manually, as an IT feature, allocating staff & resources means that staff can see which rooms are booked and which staff member booked it |

# What methods of computational thinking will be used & how will it be used to create the system?

## 1. Abstraction

Abstraction is when you focus on the most relevant and logical parts of a problem whilst ignoring unnecessary details that do not help achieve the system's main goal. In this project, abstraction will be used to identify the key areas that matter most for the hospital management system. These include the Electronic Health Record (EHR) module, the medication reminder feature, the hospital room booking system, and the secure staff login page.

These features are going to act as the core of the system because they help hospital staff carry out their daily tasks more easily, such as managing patient data, giving medication on time, and checking room availability. Focusing on these important elements makes the system simpler to design and ensures that it meets the main needs of the users, such as being accurate, easy to use, and reliable.

Some less relevant features, like optional accessibility features or multi-language settings, will still be considered but given lower priority. While they are useful for making the system more inclusive, they're not essential for the system's basic operations.

## 2. Decomposition

Decomposition is the process of breaking a large problem into smaller, more manageable parts so each section & module can be developed and tested individually. In this project, decomposition will be used to split the hospital management system into key modules that work together to form the final program.

The main parts of the system will include the Electronic Health Record (EHR) module for storing and editing patient information, the medication reminder module for sending alerts to staff, the room booking module for managing hospital spaces, and the login and authentication system for secure access. Each of these modules will be designed and tested separately before being combined into one complete program.

Using decomposition helps make the project more organised and easier to understand. It also allows errors to be found and fixed early, since each module can be tested on its own. This approach ensures modularity between different sub-components, meaning that changes can be made to one part of the system without affecting the others. This will likely make code development faster, and more efficient, helping the program meet requirements of end users

## 3. Pattern Recognition

Pattern recognition is about identifying similarities or repeated elements within a problem so that the same logic can be reused in different parts of the program. In this

project, it will be used to spot common patterns between system features and to make the code more efficient and consistent throughout.

For example, both the Electronic Health Record (EHR) and the room booking modules involve storing and retrieving data. Recognising this similarity means the same structure for data validation and error handling can be used in both modules. In the same way, the login system and medication reminder feature can share similar logic for checking user inputs and triggering specific actions at set times.

By finding these patterns early in the design stage, it becomes easier to create reusable functions/program components and avoid writing the same code multiple times. This not only saves time but also helps keep the program more reliable and easier to maintain later on.

4. **Divide and conquer**

   Divide and conquer focuses on *how* each smaller part of the project is solved rather than just identifying those parts. After the system has been broken down through decomposition, this approach is used to plan the order of development and testing. It involves taking one section at a time, building it, making sure it works correctly, and then linking it with the rest of the hospital system.

   For example, in the hospital management system, the Electronic Health Record (EHR) feature can first be developed to handle adding, editing, and deleting patient information. Once that works properly, the next part e.g. the medication reminder system can be built and tested on its own. After each module has been completed and verified, all of them are brought together to form the final working program.

   This approach helps keep development organised and reduces the risk of overwhelming errors throughout the system's source code. It also makes debugging easier, since problems can be found in a specific part of the system instead of searching through the whole program.

## 3.1.2 Stakeholders:

## 3.1.2 a) Who are the stakeholders involved & how will this software meet their needs? :

There will be a wide range of stakeholders involved in this project & would likely have an interest in the design and development of the hospital patient management system. Whether that would be directly involved (teachers, hospital staff, e.g. nurses, patients, etc) or semi-involved (Parents, Siblings, computing classmates, etc). Here is a full run-down of all the potential stakeholders involved and how they may have an impact on the project's development:
Ranking:

1 - not really useful
5 - Highly useful

| Stakeholders | Ranking | Use | Justification |
|---|---|---|---|
| Family<br>( Siblings & parents ) | 4 | • Provide feedback on all stages of the NEA e.g., analysis, design, development, evaluation.<br>• Provide feedback on code development of certain features of the system and what libraries to use in the system.<br>• They can act as simulated test users for my system, pretending to be hospital staff using my system. | My family can support me with reviewing analysis and code logic. As several of them are software engineers, they have the technical knowledge to advise me on how to implement key functions of the hospital system. However, as they have work commitments, this means that they will be unable to provide me with feedback on the project during weekdays & their opinions may also be positively biased towards the project's idea. Other than that, they can provide me with the essential knowledge needed to conduct the development of the system. |
| Teachers<br>( Computing teachers, Biology teachers ) | 4 | • Provide feedback regarding the scope of the system & what it will do in terms of features.<br>• The biology teachers may also check if the system's use of medical terminology is correct or incorrect as this system will need to be understood by real doctors.<br>• My computing teachers can also use the system to provide feedback on specific points such as the GUI | Teachers will also be a key stakeholder. When it comes to coding the software itself, my computing teachers can provide me with help & detailed feedback on how to code a complex, large-scale system, as well as review parts of the NEA write-up. I have also included biology teachers as stakeholders since they can give insights into how medical staff might interact with a system like this, helping me to shape the scope and features. However, their availability is limited to lesson times and scheduled feedback sessions. |
| Students<br>(computing classmates, Biology A-level students) | 3 | • My classmates in my computing class may also act as simulated test users, role playing as hospital staff, to use the system.<br>• They will also try to debug and find errors at random points in the system, e.g. with entering user data in the EHR, and room booking functions | While this is a lower ranking, my A-level Computing classmates & some of the A-level science students (i.e. Biology & chemistry) will help me gather primary research information/feedback on the system, as well as answers which I will gather based on a questionnaire on which I will also mock-up for this stage of the analysis. This information will help me gather quantitative & Qualitative data/feedback during development. |
| Hospital staff<br>( nurses, doctors, surgeons - simulated test users ) | 5 | • Hospital staff can use the system to simplify their day-to-day tasks such as editing patient information, and servicing patients before or after surgery.<br>• My classmates can act as the test users, role-playing as the nurses, doctors and surgeons, who will use the system.<br>• They will also give valid & honest opinions regarding the system as a whole, and what can be improved in the system's functionality. | The most important stakeholders are the intended end users of the system, i.e. the hospital staff. They represent the group this system is designed for, and so, their requirements will shape the functionality and usability of the final product. Now, for obvious reasons, I will not fake an injury just to interview the staff, and so, real hospital staff will not be directly available to me. But I will simulate their feedback using classmates, teachers & siblings, role-playing as nurses or doctors. This will allow me to test how well the system supports record-keeping, room management, and medication scheduling. Their role is vital as they provide unbiased feedback focused on how well the system meets |

| | | | |
|---|---|---|---|
| | | | its goals & performs its functions. |

## What are the accessibility requirements?

There will be a few accessibility features of the system, so that most hospital staff can use the system easily, regardless of disability, English literacy (e.g. whether they understand English), etc. Here are a few accessibility requirements for this project, to ensure that most staff can fulfil their medical duties, no matter whether or not they have disabilities.

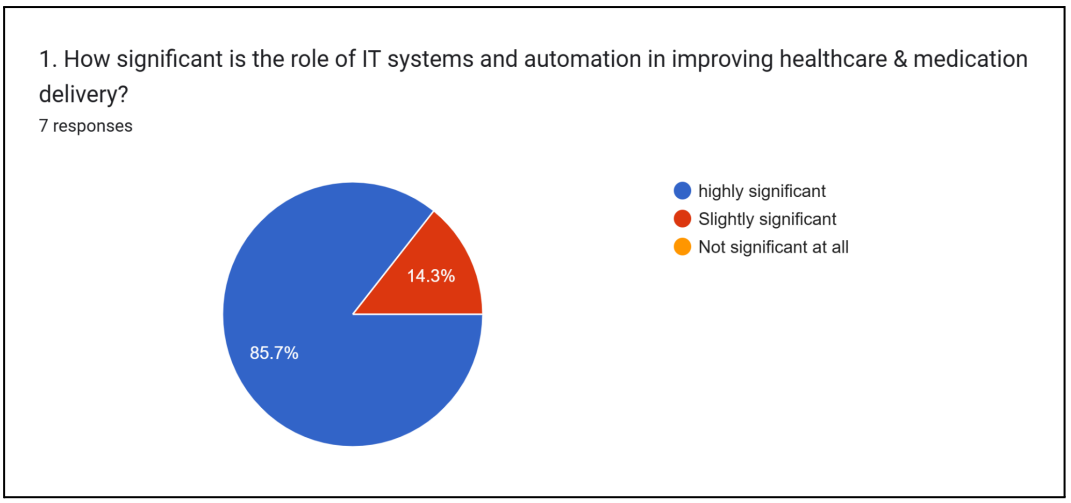
1. **Clear, easy-to-use interface (Font & contrast):**
   The system will use a clean & simple interface with a high-contrast colour scheme (e.g. dark text on a light background or vice versa). As well as a readable font size, so that any users with sight impairments or attention difficulties can easily view patient records, medication times, and room allocations, without confusion

2. **Error prevention & validation:**
   The software/system will provide clear, user-friendly error messages (e.g. "Patient ID must be unique") and prevent invalid entries when it comes to adding new patients. This supports users with cognitive difficulties by reducing confusion between patients, & preventing human errors.

3. **Multilingual support:**
   Since hospital staff may come from different language backgrounds, the system should be adaptable to allow for multilingual labels and instructions. This ensures that non-native English speakers can still use the system effectively without misinterpreting key patient or medication information. This can be changed through the settings part of the system's interface. This is also because, according to the NHS, around 28.6% of their workforce identify as Black and minority ethnic groups (BME). Therefore, due to the diversity of the healthcare industry, I am including this as a feature in my system, so all staff can use my system effectively, no matter what background or ethnicity they're from.

## Stakeholder questionnaire (Google Forms) & responses:

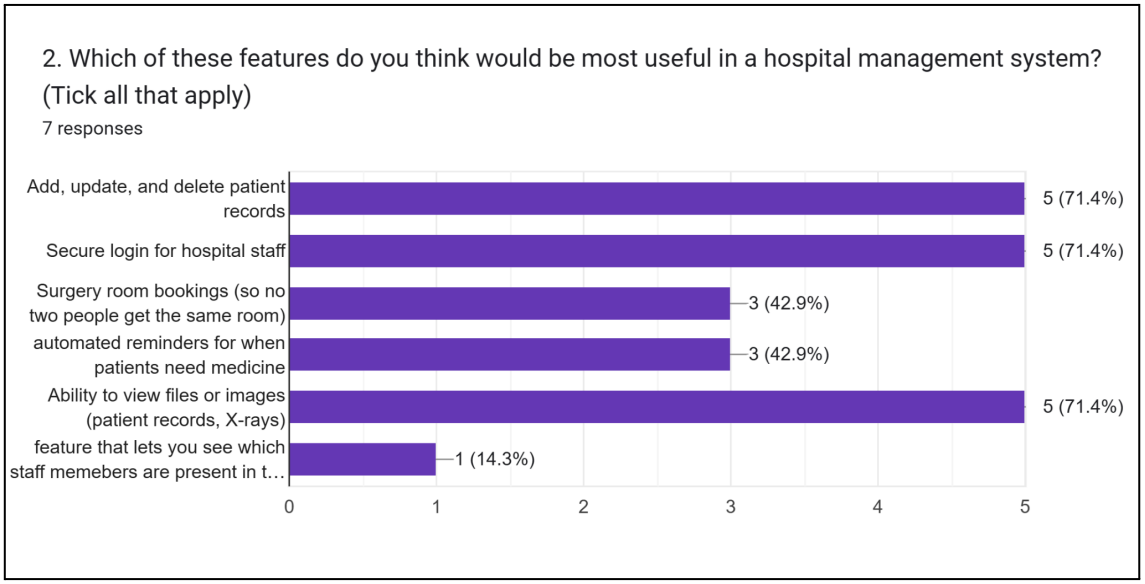
I created a stakeholder questionnaire using Google Forms and shared it with my classmates, who will act as simulated test users representing hospital staff. The purpose of this questionnaire was to gather primary research on how to improve my hospital management system, as well as to collect general views on the role of IT systems in healthcare. Below is a summary of one of the questions asked, along with the responses received.




**Question 1: How significant is the role of IT systems and automation in improving healthcare & medication delivery?**

This question highlights the vital importance of technology in shaping the field of healthcare and medicine. The results show that the overwhelming majority of respondents (85.7%) believe IT systems play a highly significant role in improving healthcare and medication delivery. The remaining responses still recognised some degree of importance, with no respondents dismissing the role of IT systems altogether.

This provides strong evidence that IT systems are widely perceived as essential in healthcare, and supports the need for a hospital management system that prioritises efficiency, accuracy, and usability.




One of the key feedback questions in my stakeholder questionnaire asked:

**"Which of these features do you think would be most useful in a hospital management system?"**

The responses are shown in the chart above. From this feedback, several important features were highlighted:

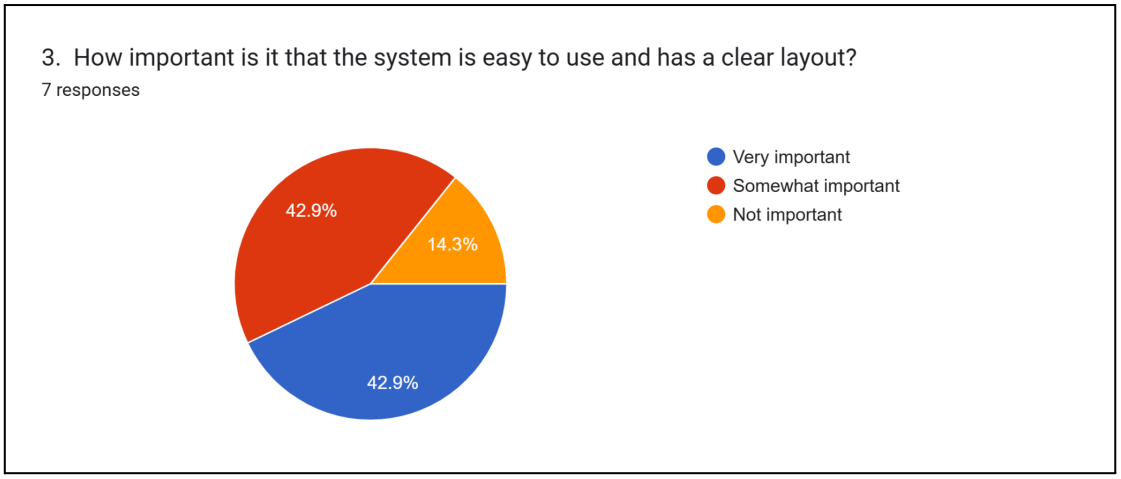
- **Add, update, and delete patient records (5 votes)**
  This was one of the most popular features. Stakeholders recognised the importance of a system that allows medical staff to maintain accurate and up-to-date patient records. This will form the foundation of my project's electronic health record (EHR) system, ensuring that medical and legal patient information can be stored and retrieved efficiently.

- **Secure login for hospital staff (5 votes)**
  Another highly rated feature was a secure login system. Stakeholders emphasised the importance of protecting patient data from unauthorised access. To address this, the system will include hashed password storage and authentication. There will also be functionality for authorised staff to add new users securely. This ensures compliance with data protection standards and safeguards sensitive, high-stakes medical information.

- **Ability to view files or images (patient records, X-rays) (5 votes)**
  Stakeholders also considered the ability to view medical images (such as X-rays, CT scans, or MRI scans) as essential. This feature will allow staff to access both graphical and text-based patient information in one place, improving usability and supporting better clinical decision-making.

- **Room booking and medication reminders (3 votes for both)**
  While slightly less popular, several respondents still identified room allocation and automated medication reminders as valuable features. These will also be incorporated to support hospital staff in managing day-to-day responsibilities.

- **Other suggested feature: Staff presence tracking (1 vote)**
  One respondent suggested an additional feature that would allow staff to see which colleagues are currently present in the hospital. This was recognised as an excellent idea for improving resource management and distributing administrative tasks more effectively. Based on this feedback, I have updated my system features (see section 3.1.1 b) to potentially include this functionality.

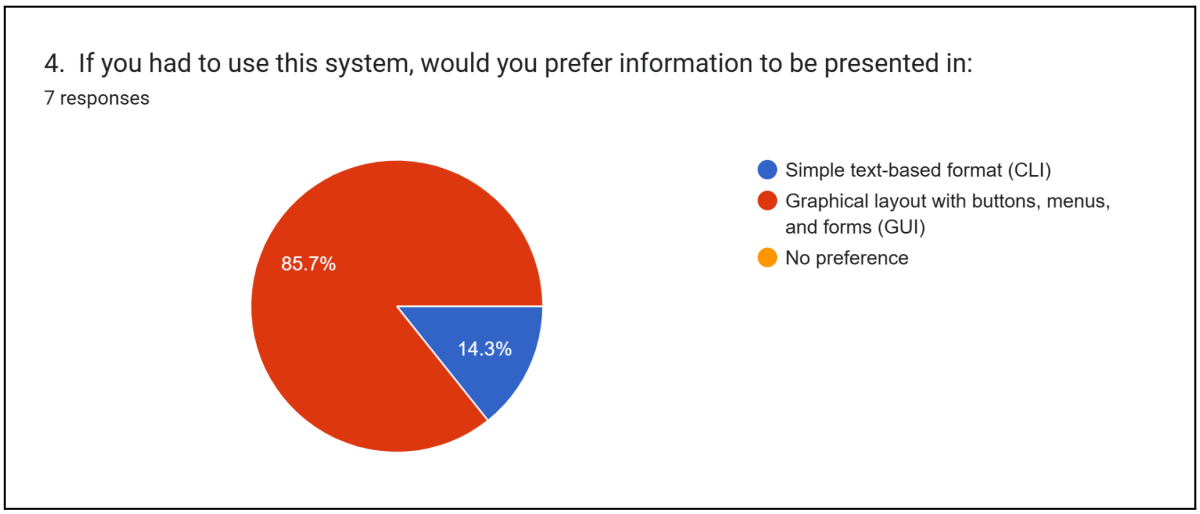



**Question 3: How important is it that the system is easy to use and has a clear layout?**

- 42.9% of respondents answered **"Very important"**
- 42.9% answered **"Somewhat important"**
- 14.3% answered **"Not important"**

The responses show that the majority of stakeholders (85.8%) consider ease of use and a clear layout to be important when interacting with the system. This highlights the need for a **minimalistic and user-friendly graphical user interface (GUI)** that enables hospital staff to quickly access patient records, view medical images (such as X-rays), book hospital rooms, check staff availability, and view medication times, for certain patients.

This feedback confirms that usability must be a priority in the system design. A simple, well-organised interface will reduce the chance of user error and improve efficiency for staff working in fast-paced hospital environments.




**Question 4: If you had to use this system, would you prefer information to be presented in:**

- **14.3%** – Simple text-based format (CLI)
- **85.7%** – Graphical layout with buttons, menus, and forms (GUI)

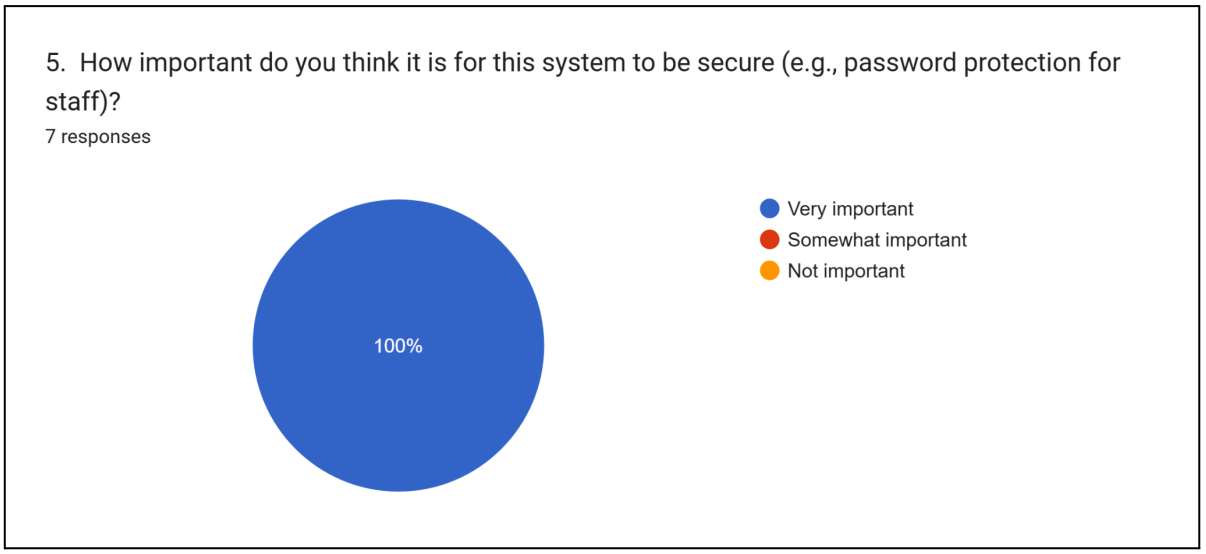
- **0%** – No preference

The results show that the overwhelming majority of respondents (85.7%) prefer a **graphical interface** over a text-based format. This feedback strongly supports the decision to design the system around a **GUI with buttons, menus, and forms**.

Based on this, the system will include a clear **side menu** that allows hospital staff to access different functions, such as:

- Patient Electronic Health Records (EHR)
- Medical images (X-rays, CT scans, MRI scans)
- Room booking
- Staff availability
- Administration tasks
- Medication schedules

This menu-based design ensures that users can easily navigate between features. For example, selecting the "EHR" option will present staff with the ability to view, edit, add, or delete patient records. This approach prioritises usability and aligns with the strong preference shown in the stakeholder feedback.




**Question 5: How important do you think it is for this system to be secure (e.g., password protection for staff)?**

- **100%** of respondents answered **"Very important"**
- **0%** answered "Somewhat important" or "**Not important**"

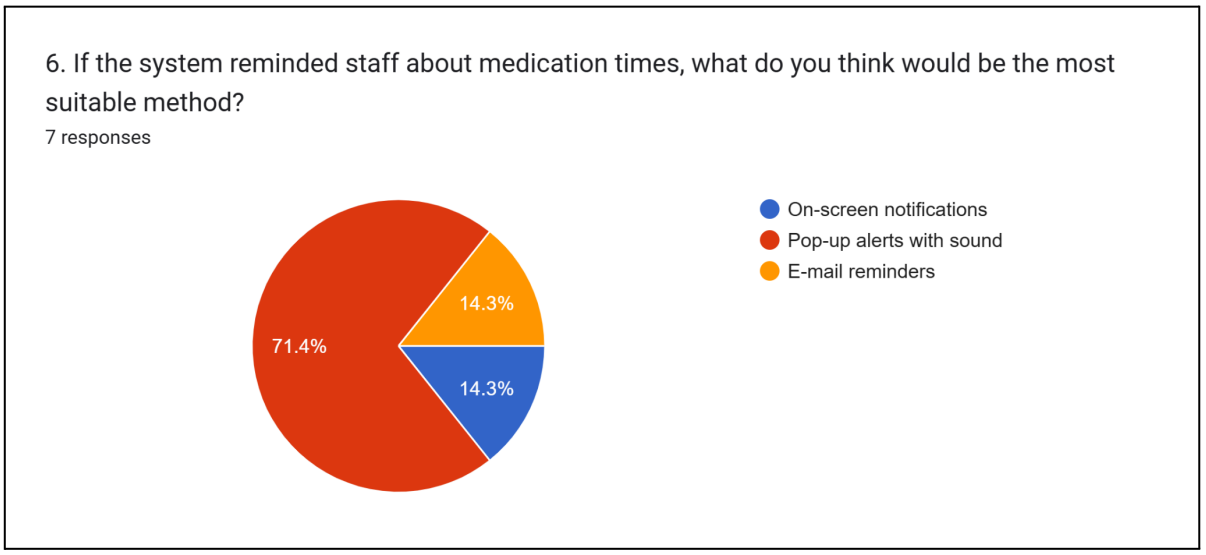
This unanimous response highlights the critical importance of **system security** in healthcare IT systems. Hospitals hold large amounts of sensitive data about patients and staff, including legal and medical information, which must be protected from unauthorised and illegal access.

Based on this feedback, the system will implement **secure staff login using hashed passwords**. Instead of storing passwords in plain text, they will be stored in a hashed form within a hash

table. When a user attempts to log in, the entered password will be hashed and compared against the stored hash value. If the values match, then access will be granted. If not, access will be denied.

This approach ensures that even if the database were accessed, the actual passwords would remain protected. Furthermore, the use of hashing prevents attackers from exploiting similar-looking passwords, since even small differences in input produce completely different hash outputs.

By following this method, the system meets both stakeholder expectations and standard practices in secure software design.



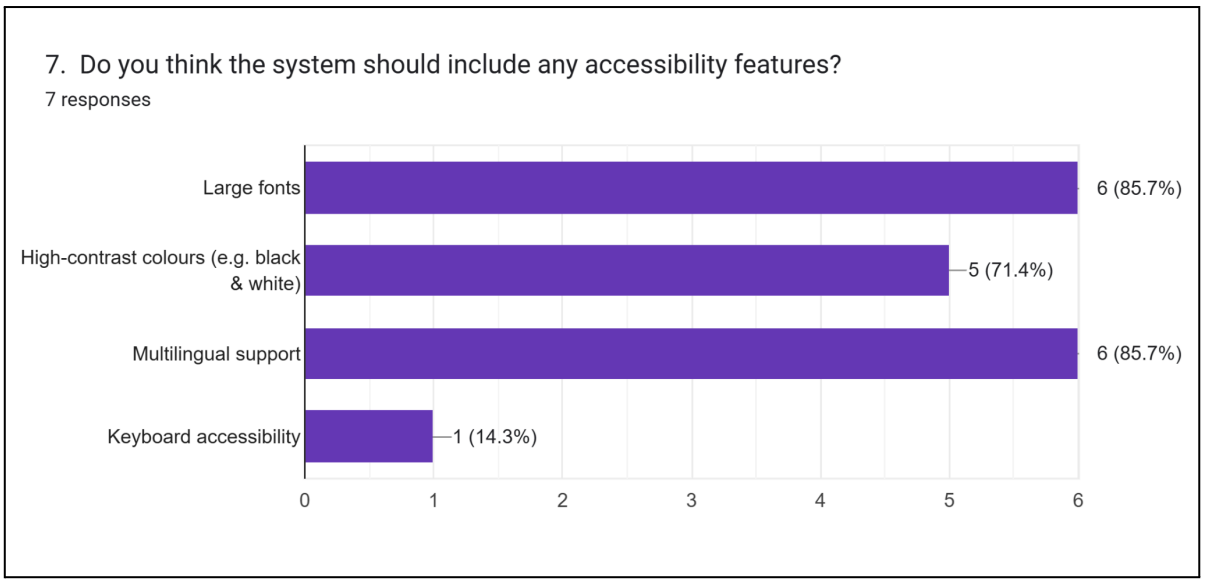

**Question 6: If the system reminded staff about medication times, what do you think would be the most suitable method?**

- **71.4%** – Pop-up alerts with sound
- **14.3%** – On-screen notifications
- **14.3%** – Email reminders

The majority of respondents (71.4%) preferred **pop-up alerts with sound** as the best method for medication reminders. This option ensures that staff both see and hear the alert, making it less likely that medication times are missed.

By contrast, on-screen notifications and email reminders rely on staff actively checking their devices, which increases the risk of overlooking a reminder. Therefore, the feedback indicates that **pop-up alerts with sound** provide the most reliable and effective solution, as they do not depend solely on visual attention but also make use of audible cues to capture staff awareness.

Therefore, this feature will be implemented into the system design to ensure that medication schedules are managed accurately and efficiently.




**Question 7: Do you think the system should include any accessibility features?**

- **85.7%** – Large fonts
- **71.4%** – High-contrast colours
- **85.7%** – Multilingual support
- **14.3%** – Keyboard accessibility

The majority of respondents considered **large fonts** and **multilingual support** to be the most important accessibility features (both selected by 85.7% of respondents). These features will ensure that staff with visual impairments or attention difficulties can clearly view patient information, medication times, and room allocations, while multilingual support will allow staff from diverse language backgrounds to use the system without confusion or the risk of misinterpretation.

**High-contrast colours** (selected by 71.4%) were also seen as important, as they improve readability and reduce eye strain when viewing medical records and schedules.

A smaller proportion of respondents (14.3%) selected **keyboard accessibility**. While this can support users who cannot operate a mouse, it is less relevant for the target audience/simulated test users (classmates) of this project. Therefore, although this option will be noted, it will not be prioritised in the system's core design.

By including large fonts, high-contrast colours, and multilingual support, the system will be more inclusive and accessible to a wide range of potential users, aligning with both stakeholder expectations and good design practice.

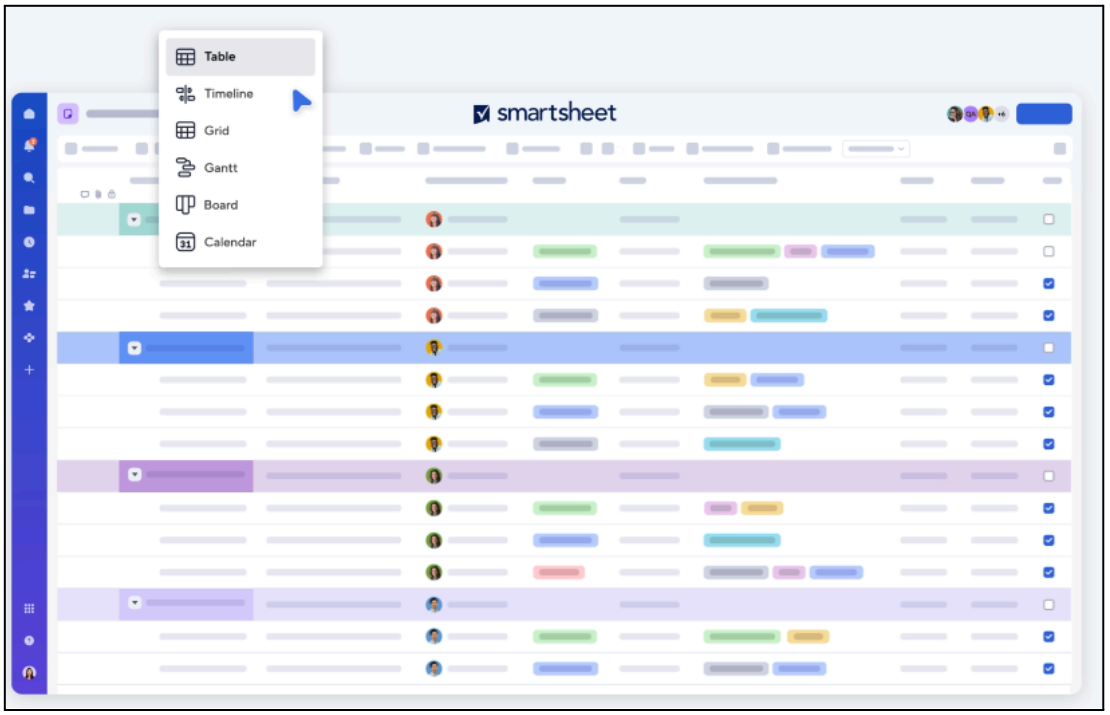
## END OF QUESTIONNAIRE

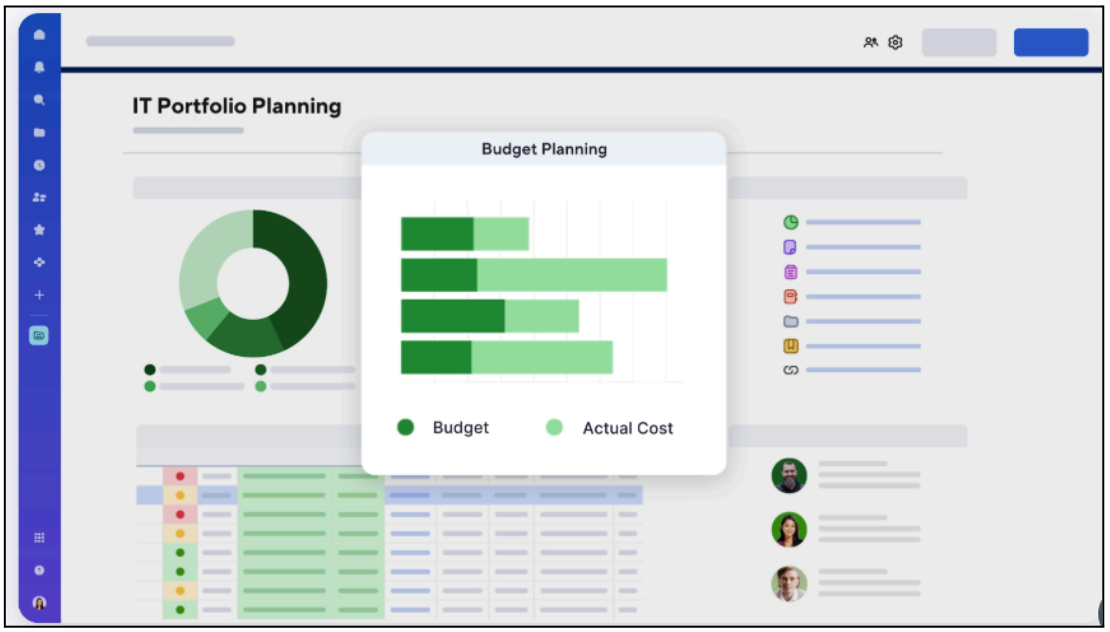
## 3.1.3 Research the problem:

### 3.1.3 a) What problems and solutions are out there that are similar to my project?:

There are many similar solutions out there that are closely related to my project. Here are a few that stuck out to me the most! I also have to mention that I got these programs from a website called PostGrid.com, with the following URL: (https://www.postgrid.co.uk/top-hospital-management-software/) for the 10 best hospital management software programs in 2024. I have also included different tables, which address the differences and similarities between my medical system and their medical system.

1. **Smartsheet:**






Whilst this is an application that can be used for general purposes & most projects, at its core, this is a spreadsheet management system, meaning that it will likely be well suited for storing patient records and information. This allows hospital staff to manage, track & report patient records, similar to a standard EHR system (**Electronic Health Records**) or EPR (**Electronic Patient Records**) system. However, unlike my system, Smartsheet allows staff to collaborate on tasks easily and hassle-free. Additionally, this system allows staff to manage, store and share PHI (**Protected Health Information**), securely. This helps avoid regulatory and legal issues. This type of system is more suited to small clinics or GP practices, where teams are smaller and medical projects are less complex.

| Similarities | Differences |
|---|---|
| • Has a built-in EHR system to store and manage patient records.<br>• Room booking management, allowing staff to see which rooms are available or in use. | • Offers inventory management, allowing staff to monitor stock levels and reorder medication when necessary.<br>• Includes financial functionality, such as processing payments and managing insurance plans. |

## 2. athenaOne:

athenaOne Calendar Patients Claims Financials Reports Quality Support

MC Michael CADY
46yo M | 02-14-1977 | #1234 | E#1330

Contact: Mobile 555-555-5555
Appointments: Next Follow up Exam 04-07-24
Provider: Usual John Garcia, MD

Established Patient | John Garcia, MD

Go to Intake

Find, Allergies, Problems, Meds, Vaccines, Vitals, Results, Visits, History, Quality

**Allergies**
- sulfa (sulfonamide antibiotics)
- peanut
- penicillins

**Problems**
- mixed hyperlipidemia
- diabetes melitus
- essential hypertension

**Medications**
- alprazolam
- atorvastatin
- lisinopril
- medrol
- mupirocin
- naprosyn
- tadalafil
- zithromax
- zolpidem

**Surgical History**
- hernia repair: 12-0202019

**Last Visit with Internal Medicine** | Recent Activity

follow up 15, 08-04-2023
Performed by Irene Medina, MD, Internal Medicine, 909-756-1234

Discussed importance of healthy diet, routine exercise and medication compliance. Reviewed recent lipid results. Discussed treatment options including starting a statin. Patient wants to continue to try lifestyle modification before considering medication. Will recheck lipids prior to next visit.

1. **Mixed hyperlipidemia**
E78.2: Mixed hyperlipidemia
LIPID PANEL AND CHOL/HDL RATIO

2. **Diabetes mellitus**
E11.9: Type 2 diabetes mellitus without complications
HEMOGLOBIN A1C-001453-P

3. **Essential hypertension**
I10. Essential (primary) hypersension
lisinopril 2.5 mg tablet -
Take 1 tablet(s) every day by oral route as directed for 30 days. Qty 30 tablet(s)
Refills: 0 Pharmacy: CVS/PHARMACY #8436

**Outstanding Orders**
hemoglobin A1C-001453-P (submitted)

## Transactions History

**Claim #441678**

Transactions Overview: 5 Charges

| | Transaction Type | Claim DENIED | CPT 99213 | CPT 99051 DENIED | CPT 99214 | CPT 30776 | CPT 30778 |
|---|---|---|---|---|---|---|---|
| Billed to Johns Ho... (Primary) Created on 05/03/2024, mrozmiarek | Charge [CLOSED] | $300.00 | $206.00 | $54.00 | $40.00 | $0.00 | $0.00 |
| ERA from Johns H... (Primary) #29695860 \| Replicated EOB Posted on 05/14/2024, ATHENA (ERA) | Payer payment (ACH234782315) | -$100.86 | -$75.86 | - | -$25.00 | - | - |
| | Contractual Adjustment (CO45) | -$145.14 | -$130.14 | - | -$15.00 | - | - |
| | Denial (CO288) | - | - | $0.00 | - | - | - |
| | **Outstanding** | $54.00 | $0.00 | $54.00 | $0.00 | $0.00 | $0.00 |

Transaction Details by Charge : 5 Charges

Procedure code: 99213 | 99051 | 99214 | 30076 | 30078

| Date | Procedure | Diagnoses | Units | Department | Charge |
|---|---|---|---|---|---|
| 05-03-2024 | 99051 | D225, L814, Z872 | 1 | MAIN OFFICE | $54.00 |

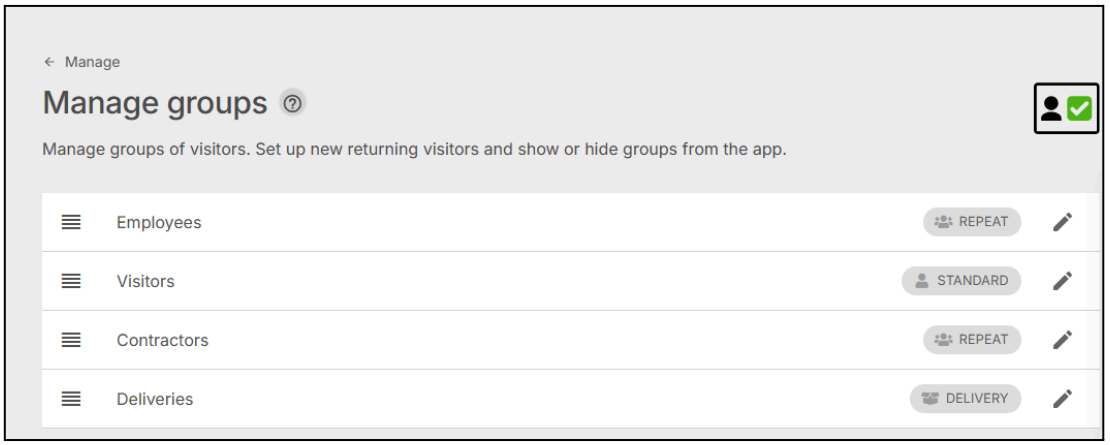
Show voided transactions (1)

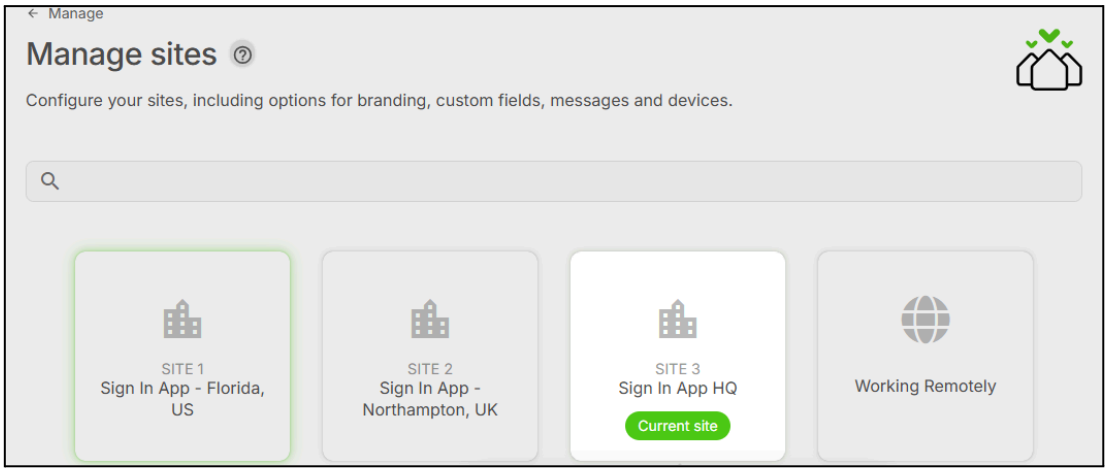
| | Transaction Type | Primary Insurance | Secondary Insurance | Patient | Action |
|---|---|---|---|---|---|
| Billed to Johns Ho... (Primary) Created on 05/03/2024, mrozmiarek | Charge [CLOSED] | $54.00 | | | |
| ERA from Johns H... (Primary) #29695860 \| Replicated EOB | Denial(CO288) Referral is absent or missing | $0.00 | | | |

athenaOne is a multi-purpose hospital management program designed to handle a wide range of tasks, such as summarising administrative work and organising patient care plans. One of its standout features is the use of advanced analytics, which enables the system to learn from a patient's medical history and recommend relevant prescriptions or treatments. This data-driven approach enables the provision of highly effective, tailored healthcare solutions for patients.

| Similarities | Differences |
| --- | --- |
| • Supports hospital staff in managing administrative and medical tasks.<br>• Provides an EHR system that enables doctors and staff to store accurate and up-to-date patient records. | • Includes payroll features, allowing staff to receive payments and wages directly through the system.<br>• Tracks physician orders, such as medication refills and logistical requests.<br>• Connects with a large network of Athena-owned services (e.g. **AthenaClinicals, AthenaServices** ) |

## 3. Sign-In Scheduling:




This is an appointment management system designed to improve patient experience and satisfaction. This scheduling system uses advanced analytics to predict high demand for medical services, helping hospitals and clinics avoid overcrowding by balancing appointment bookings. By identifying both peak and quiet periods, the system allows staff to allocate resources more effectively, reduce waiting times, and enhance the hospital's overall efficiency.

| Similarities | Differences |
|---|---|
| • Is able to send automated medication reminders for staff and patients, using E-mail or SMS, instead of sound alerts.<br>• Allows hospital staff to manage patient records on their EHR. | • Includes predictive analytics to forecast service demand for hospitals.<br>• Offers financial management features to track bills and hospital expenses. |

## What features from these systems can be included in my project?

One feature that could be incorporated into my project is **inventory management**, specifically the ability to track the stock levels of certain medications. Since my planned software project already includes elements of resource management and staff allocation, extending this to cover supply levels of medication and enter medication info would add value to hospital staff. This functionality would help hospital staff identify when medicines are running low or are close to being out of stock, ensuring that supplies can be replenished before shortages occur.

## 3.1.3 b) What are the essential features of the hospital patient management system?:

My hospital patient management system will include a range of essential features designed to support hospital staff in managing day-to-day administrative and clinical tasks. These tasks include administering medication, updating patient medical records, and viewing medical images.

Below is an outline of the **core essential features** that will be implemented. It is important to note that this list focuses only on the essential functions of the system. Additional semi-essential features are listed separately at the bottom of the table as a reference.

| Feature | Description | Justification |
|---|---|---|
| Built-in EHR for staff to manage patient records | This feature will allow staff to add, update, view and delete patients' medical information. | Staff must stay updated on patients' medical issues (e.g., symptoms, medication, contact information). A built-in EHR ensures that records are accurate, accessible, and centralised, helping staff provide effective care. |
| Password and username protection | Enables staff accounts (e.g., username and password) to be securely protected using a hash | Protecting sensitive patient data is essential. Storing hashed passwords reduces the risk of breaches if unauthorised users attempt to access the |

| | | |
|---|---|---|
| | function and a table to store hashed credentials. | system. This ensures that confidential medical and legal information remains secure. |
| Automated medication reminders | Alerts staff (with sound notifications) when it is time for specific patients to receive medication | This feature ensures patients are given medication on time, improving treatment accuracy. It removes the need for staff to manually track schedules, reducing human error and improving patient safety. |
| Room booking management | Allows hospital staff to book available rooms (e.g., surgery rooms) and view which rooms are in use, by whom, and for how long. | Prevents double-booking, reducing scheduling conflicts, delays, and confusion between staff and patients. |
| View medical images | Enables surgeons and doctors to view medical images such as CT, MRI, and X-ray scans within the system. Each image is linked to the relevant patient record. | Provides staff with a clear way to visualise symptoms and injuries, supporting more accurate treatment decisions. |
| Graphical User interface | Provides a simple, clear interface that integrates all system features into one accessible layout for hospital staff. | A GUI ensures ease of use. Unlike a command-line interface, it does not require advanced technical knowledge, allowing staff to navigate features efficiently. |

## What are the semi-essential features?

Although the following features are not critical for the system to function, they would add significant value by improving usability and efficiency. These features are therefore considered **semi-essential**, as they enhance the system for hospital staff (or simulated role-playing classmates).

1. **View staff availability:**
   This feature will allow hospital staff to check whether colleagues are available or currently engaged in medical duties. It can also be integrated into the room booking

page, showing which staff members have booked rooms, the times they are occupied, and when they will next be free. This would improve coordination between staff.

2. **View stock levels:**
   This feature will enable staff to view the stock levels of medication within the hospital's inventory. Staff can input the medication details into the system, which will then display whether supplies are running low or need replenishing. This helps prevent shortages and ensures essential medicines are always available.

3. **Staff login:**
   This feature will allow staff to securely log in to the system using a username and password. Login details will be protected using a hash function, ensuring that sensitive data is not stored in plain text. While a secure login is already considered an essential requirement, this semi-essential extension covers additional implementation details such as the hashing process and storage of encrypted credentials.

## 3.1.3 c) What are the limitations of my hospital program?:

There are a few features that I decided not to include in my project, as they would either over-complicate the scope or are less relevant to the needs of my stakeholders. Below is a summary of the excluded features and the reasons for their absence.

| Limitation | Reasoning |
|---|---|
| Keyboard-exclusive accessibility (system can only be used with the keyboard) | This feature was excluded because all of the current functions (e.g., viewing patient records, booking rooms) already account for standard hardware requirements such as a mouse and keyboard. None of my stakeholders or simulated test users identified this as a necessary requirement, so it is not as important as other features. |
| Payment and financial features (e.g. handling customer payments or insurance) | While this feature could be useful in some hospital systems, I decided not to include it as it would over-complicate the scope of the project. The system is focused on supporting hospital staff with medical and administrative tasks, rather than finance. Similar systems (e.g., athenaOne, Sign-In Scheduling, Smartsheet) often include financial tools, but for the purposes of my project, this functionality is not essential. |
| Multilingual support (supporting different languages) | Although this feature is valuable in real-world hospital systems, I chose to exclude it as it would add unnecessary scope and complexity to my project. My stakeholders and simulated test users already read and understand English fluently, so this feature was not considered a priority. Excluding it allows me to place greater emphasis on developing and testing the system's core medical features. |

## 3.1.4 Specify the proposed solution:

### 3.1.4 a) What are the hardware and software requirements of the system?:

My system will have a few hardware components, as well as a few software components, to ensure the system runs smoothly and properly. Below, I have written a summary rundown of all the requirements (**stakeholder, functional, hardware/software**) necessary for the system to function!

| Requirement | Category | Explanation |
|---|---|---|
| **Any python IDE, capable of running python 3.0** | Hardware/software | An IDE is required to run and test the system's source code. This can be achieved using platforms such as Visual Studio, PyCharm, or Replit. |
| **Keyboard** | Hardware/software | Used by staff to enter patient details into the EHR (e.g., medical history, medication) as well as staff details (e.g., staff name, occupation). |
| **Mouse** | Hardware/software | Allows staff to interact with the GUI by selecting tabs, menus, and buttons (e.g., Room Bookings, EHR, medication times, staff availability, medical images). This enables efficient navigation of the system. |
| **Speakers / sound card** | Hardware/software | Required for the medication reminder feature, as the system will generate sound alerts to notify staff when medication needs to be administered and given to patients. |
| **Graphics card/GPU** | Hardware/software | Required for displaying high-quality medical images such as CT scans, X-rays, MRI scans, and ultrasound images. This ensures hospital staff can clearly view and interpret diagnostic information. |
| **GUI (Graphical User Interface)** | Stakeholder/functional | Acts as the central interface for the system, combining all features into a single, simple layout. The GUI ensures that staff can easily navigate the system without needing advanced technical knowledge. |
| **Password validation using a hash table** | functional | Ensures that login credentials are securely stored and validated. Passwords are hashed before being saved, and when re-entered, the hash value is |

| | | |
|---|---|---|
| | | compared against the stored version. This prevents unauthorised access to patient records and maintains system security. |

## 3.1.4 b) What are the success criteria for the hospital management system

My system will include a success criteria (wish list) to measure whether it meets all the planned requirements. The table below outlines the criteria, the evidence that will demonstrate success, and the justification for including each requirement:

| criteria | Evidence | Justification |
|---|---|---|
| Staff login page grants access within 5-10 seconds if correct details are entered | Screenshot and timed screen recording of the login process | Ensures staff can access the system quickly with minimal delay, improving efficiency. |
| Authorisation for new staff to enter the system, using the details of an existing staff member. | Screenshot of the new staff authorisation page, requiring an existing staff member's credentials. | Guarantees only authorised users gain access. A secure authorisation step protects sensitive data and allows safe addition of new staff. |
| GUI for navigating the different system options | Screenshot of GUI layout, displaying all options clearly. | Provides staff with a clean, user-friendly layout, making it easy to navigate between features and reducing training needs. |
| EHR system for staff to enter patient details (medication, legal information, medical history, etc) | Screenshots and stakeholder testing | Enables staff to keep patient records digitally instead of storing them physically, ensuring records are accurate, up-to-date, and easily accessible. |
| View medication times for patients | Screen recording of medication reminders and sound alerts. | Sound-based alarms reduce the risk of staff missing medication times, improving accuracy and reliability of medication delivery. |
| View medical images (X-rays, Sonar, CT scans, MRI, Ultrasound, etc ) | Screenshot of the GUI displaying a medical image. | Allows staff to visually analyse patient symptoms and injuries. A dedicated tab for medical images within patient records will likely ensure clarity and efficiency for hospital staff. |
| | | Helps staff see which colleagues are free to assist with duties such as |

| | | |
|---|---|---|
| View staff availability and current staff tasks | Screenshot of the staff availability tab/menu. | distributing medication, while others can focus on tasks like updating records. Therefore, improves workload management. |
| View stock level of medication | Screenshots of medication stock entries and system registration of details. | Ensures staff can monitor medication levels and reorder when supplies are low. This prevents shortages and guarantees patients always receive medication on time. |

# End of analysis